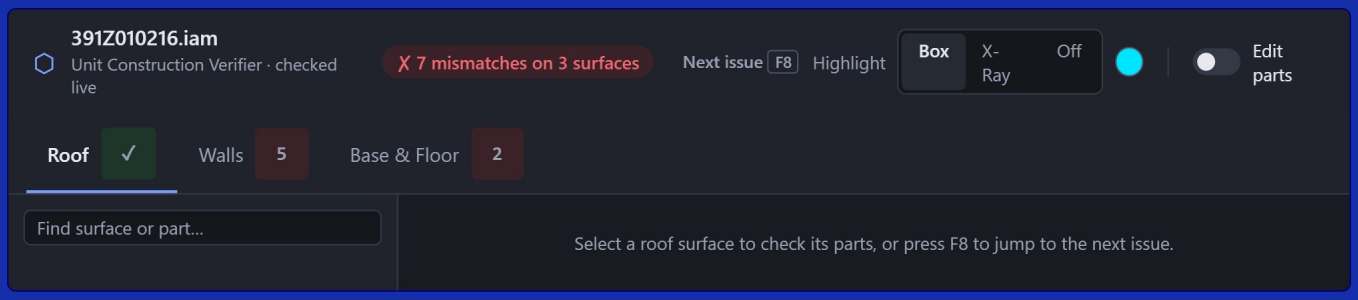
QA TOOLS FOR AUTODESK INVENTOR 2020

Unit Construction Verifier

User guide: setting it up, checking a unit, fixing parts, and finding answers

Version 1.0.40 · November 2026



The verifier checks that every part of an air handling unit modelled in Inventor is made of the gauge and material the order calls for, and helps you fix the ones that are not. No experience with the tool is needed to follow this guide.

Set it up
Install the add-in and find it in Inventor.
Pages 4–6

Check a unit
Read the window and find every mismatch.
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Fix and upload
Correct parts, regenerate and upload to MOM.
Pages 13–19

Look it up
Troubleshooting, terms and shortcuts.
Pages 20–21

For internal use

Johnson Controls

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How to read this guide

Re-check unit Bold words are exactly what you see on screen: a button, a tab or a menu item.

1 Orange numbered circles in a picture match the numbered notes next to it.

1 Blue numbered squares are steps. Do them in order.

QA Tools › Construction The › sign means “then click”.

GOOD TO KNOW

Grey boxes hold tips you can skip on a first read.

BE CAREFUL

Orange boxes warn about something that changes files or cannot be undone.

The pictures are drawn from the verifier's real layout, colours and labels. Unit, surface and part numbers in them are examples; yours will differ.

Johnson Controls

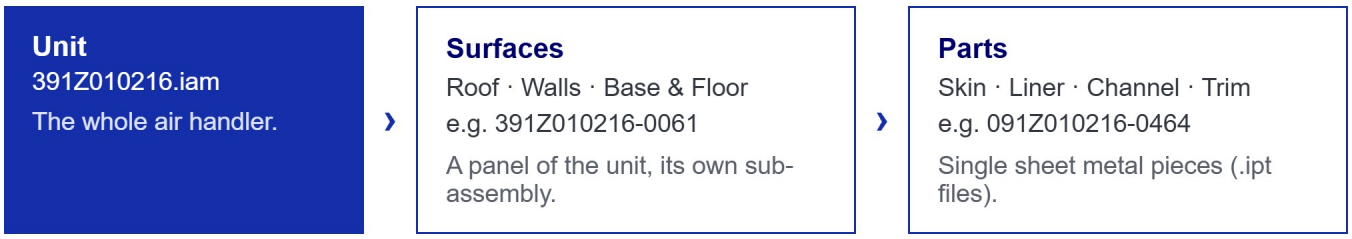
CHAPTER 1

What the verifier does

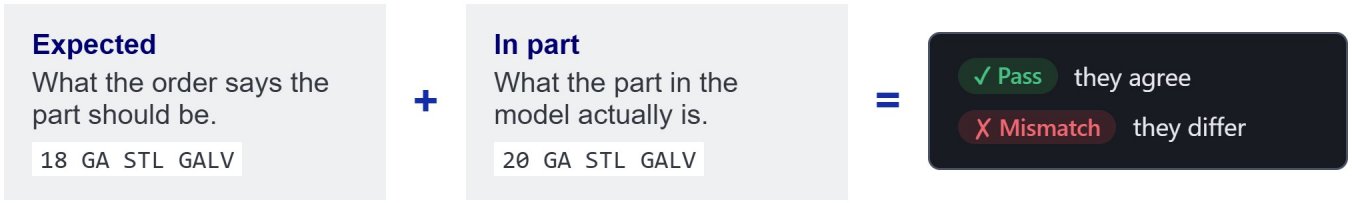
Every air handling unit is modelled in Inventor as one large assembly file (an **.iam**). The unit is split into **surfaces**: roof panels, walls, and the base and floor. Each surface is built from **parts** such as the outside skin, the inside liner, the channels that frame it, and trim.

The order for the unit says what those parts must be made of, for example “18 gauge galvanised steel” for a wall's outside skin. The verifier reads that from the unit itself, then looks at every part in the model and tells you whether it matches.

How a unit is organised



How a part is checked



What it will and won't do

It will

· Check every covered part on every surface, live, as soon as it opens.

· Point to each mismatch and show the part in the model.

· Change parts for you, but only after you review and press **Write**.

· Run YCShell's Regenerate and Upload on all changed surfaces in one go.

It won't

· Change any file just by being open or by checking.

· Write a staged change when you close it.

· Replace YCShell or MOM. It presses their buttons for you.

· Check parts that no rule covers. Those are left out of the list.

GOOD TO KNOW

Checking is safe. You can open the verifier on any unit, look around and close it without changing anything.

CHAPTER 2

Install it on your PC

Installing takes about two minutes. You only copy a few files into your own Inventor folder; nothing needs building and you do not need administrator rights.

Before you start

You need	Why
Autodesk Inventor 2020 on a Windows PC	The verifier runs inside Inventor.
The file `UnitConstructionVerifier_Dist.zip`	It holds the add-in. Get it from: `P:\Detailing\Detailing Tools`
Only for **Regenerate** and **Upload**: the JCI MOM (YCShell) add-in, and a login to the MOM tools	The verifier drives YCShell for these steps (chapter 7).
Only for **Edit parts**: the part files checked out of Vault	Parts that are checked in cannot be saved.

Get the files ready

1. **Save your work and close Inventor.** The installer can't replace the add-in while Inventor is using it.
2. Copy `UnitConstructionVerifier_Dist.zip` to your **Downloads** folder. Right-click it › **Extract All…** › **Extract**. Don't run anything from inside the zip itself.
3. Open the new folder that appears. It should hold these seven files. If any are missing, get a fresh copy of the zip.

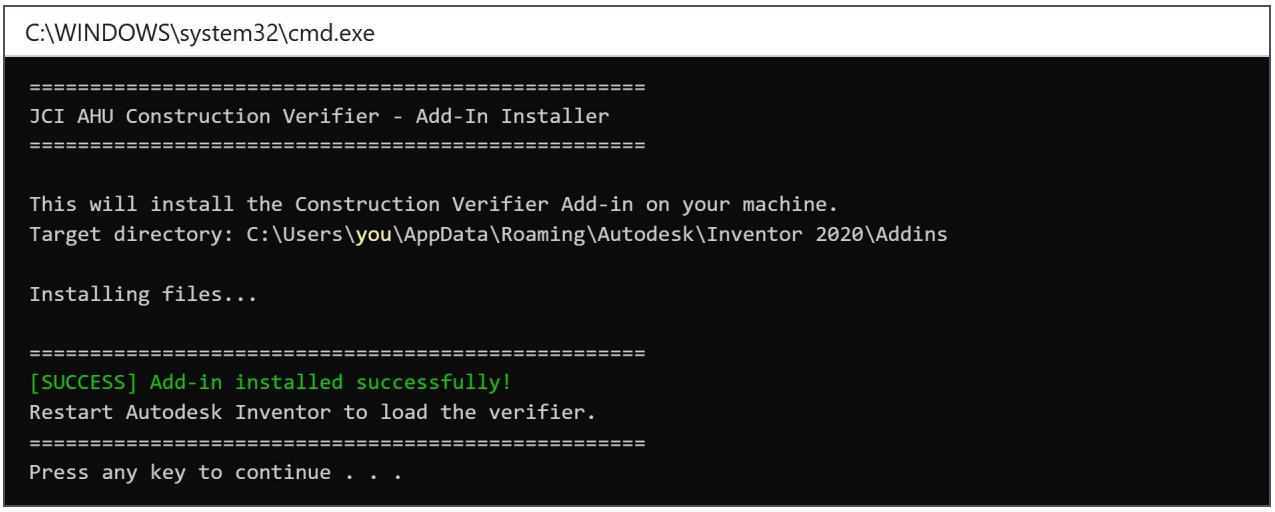
Name	What it is	Size
install.bat	The installer. Run this.	3 KB
UnitConstructionVerifier.dll	The add-in itself	448 KB
UnitConstructionVerifier.addin	Tells Inventor to load it	1 KB
Newtonsoft.Json.dll	A helper it needs	696 KB
materials_config.json	Checking rules	17 KB
materials_thickness_map.json	Gauge thicknesses	4 KB
version.txt	The version number	1 KB

Contents of the extracted zip (version 1.0.40)

Run the installer

4 Double-click **install.bat**. A black window opens. If Windows asks whether you want to run the file, choose **Run**.

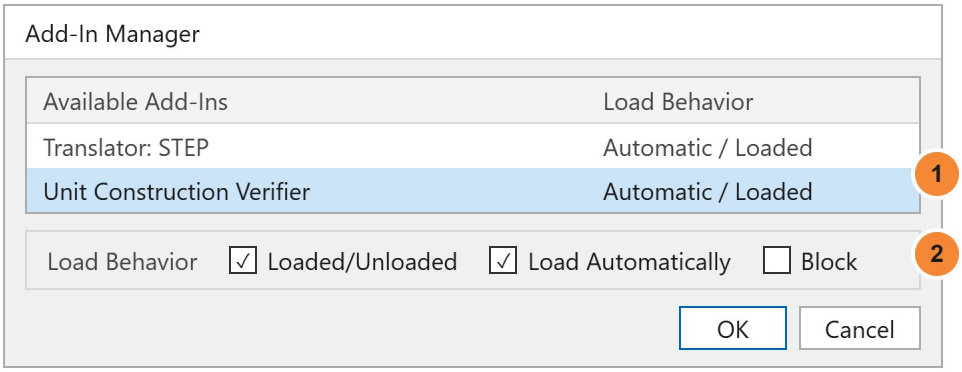
5 Wait for **[SUCCESS] Add-in installed successfully!**, then press any key to close the window. If it shows **[WARNING]** or **[ERROR]** instead, see page 20.



What a successful install looks like. “you” is your Windows user name.

Turn it on (first time only)

6 **Start Inventor**, open any assembly, and click **Tools › Add-Ins**. In the list, click **Unit Construction Verifier**. Under **Load Behavior**, tick **Loaded/Unloaded** and **Load Automatically**; if **Block** is ticked, untick it. Click **OK**.



Simplified Add-In Manager. **1** Select it. **2** Tick as shown. The **QA Tools** tab then appears (page 6), and loads by itself from now on.

NEW VERSIONS

They install themselves. When Inventor starts, the verifier copies any newer version from the same folder and shows a notice; it is used from the next start.

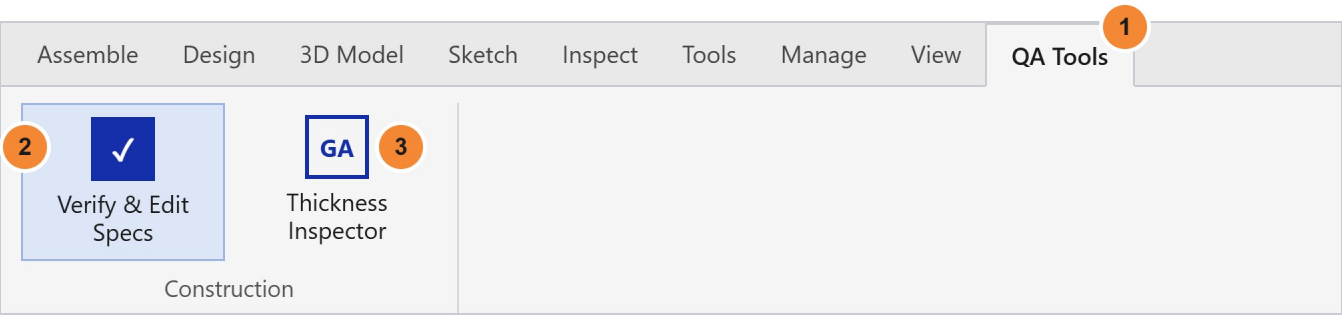
REMOVING IT

To pause it, untick **Load Automatically** in step 6. To remove it, delete its files from the **Target directory** folder.

CHAPTER 3

Open it in Inventor

1. Open the unit's **main assembly** (the whole unit's `.iam`, not one surface or part).
2. Click the **QA Tools** tab at the top of Inventor.
3. In the **Construction** panel, click **Verify & Edit Specs**. The verifier opens and checks the unit straight away.



The QA Tools tab. Button icons are drawn for this guide and may look different in Inventor.

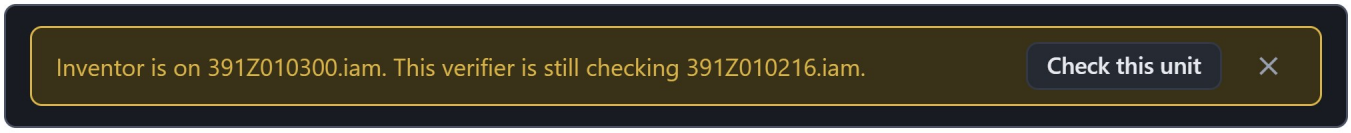
1. **QA Tools** tab. Shown when an assembly or a part is open.
2. **Verify & Edit Specs** opens the verifier. Assemblies only.
3. **Thickness Inspector** shows a part's gauge on hover (page 19).

Where the window goes

The verifier sits beside the model. You can keep rotating and zooming in Inventor while it is open. Click **Dock** at the right end of its header to put it in a panel inside Inventor instead; drag the panel's title bar to move it to the left or bottom. **Pop out** puts it back in its own window. It opens the next unit the way you left it.

When the window is narrow, the layout switches to one column: surfaces on top, and **Expected construction** folded away until you click it.

If Inventor switches to a different unit, a yellow bar appears:

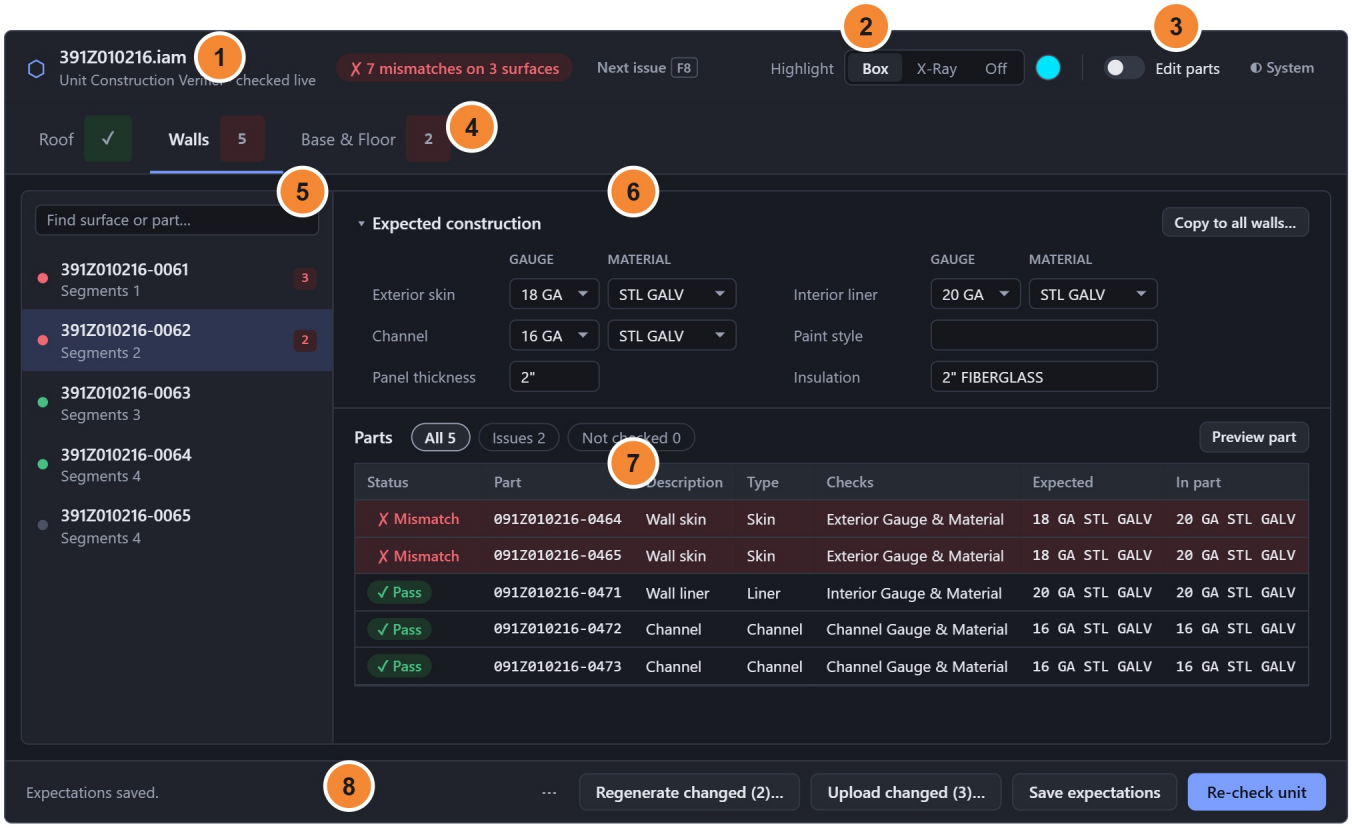


Check this unit closes the current check (asking first if you have unsaved changes) and checks the unit Inventor is on. ✕ keeps checking the first unit. Opening one of the unit's own surfaces does not count as a different unit.

CHAPTER 4

A tour of the window

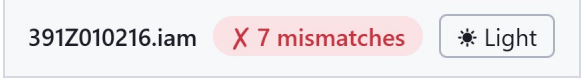
The verifier has eight areas. Read it top to bottom: the header gives the overall answer, the tabs and list narrow it to one surface, and the parts list shows each part's result.



1. **Unit and result.** Which unit is open and how many parts don't match. Page 8.
2. **Highlight.** How the selected surface or part is marked in the model. Page 8.
3. **Edit parts.** Turns on fixing parts. Off by default. Page 13.
4. **Surface tabs.** Roof, Walls, and Base & Floor, each with its own count. Page 9.
5. **Surface list.** Every surface on this tab, with a coloured dot and a count of problems. Page 9.
6. **Expected construction.** What this surface's parts should be. Page 10.
7. **Parts list.** One row per checked part, with its result. Page 11.
8. **Footer.** Save, re-check, regenerate, upload and more tools. Pages 12–18.

Light or dark

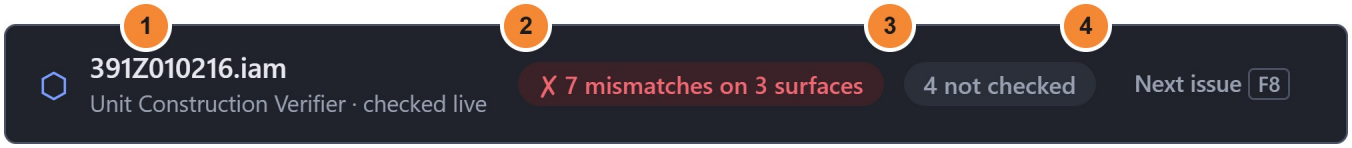
The ◐ **System** button in the header picks the colours. **System**, the starting choice, matches Windows' light or dark mode and changes when Windows does. Each click moves on: ☀ **Light**, ☾ **Dark**, then back to **System**. Your choice is remembered. The pictures in this guide show the dark theme.



The header in the light theme.

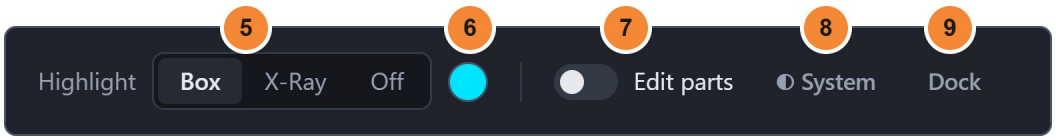
The header

The strip across the top answers “is this unit right?” and updates by itself.



1. **Unit name.** The unit this window is checking.
2. **Result.** The headline answer. See the three states below.
3. **Not checked.** Parts a rule covers that could not be checked. The parts list says why (page 11).
4. **Next issue** (or the **F8** key) jumps to the next surface with a mismatch and selects its first bad part.

Result shows	It means
✓ All 214 checked parts match	Every part that was checked is right. Nothing to do.
✗ 7 mismatches on 3 surfaces	Some parts differ from the order. Press **F8** to visit them one by one.
Not scanned	The unit could not be read, so nothing was checked. Make sure the unit's main assembly is open, then re-open the verifier.

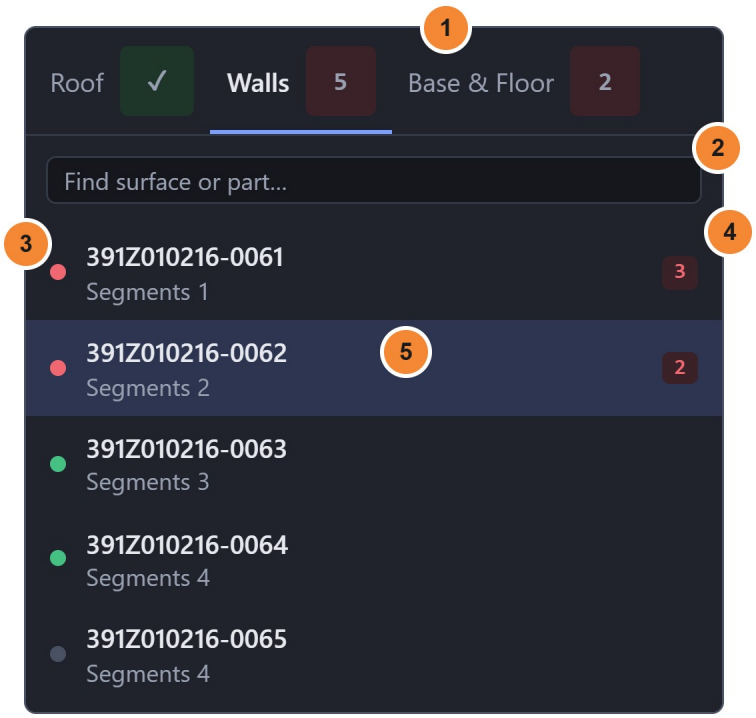


5. **Highlight** picks how the selected surface or part is shown in the model. See below.
6. **Colour dot.** Click to choose the highlight colour.
7. **Edit parts** lets you change parts. Leave it off just to check. Chapter 6.
8. **Theme.** Click for light or dark colours. Page 7.
9. **Dock** / **Pop out** moves the verifier into Inventor or back to its own window.

Highlight	What you see in the model
Box	A coloured box around the selected surface or part. This is the default.
X-Ray	A detailed outline of the selected part's edges, so you can see it inside its surface.
Off	No verifier highlight; Inventor's green selection comes back until you pick Box or X-Ray.

Surface tabs and the surface list

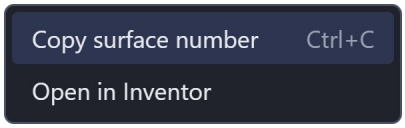
Surfaces are grouped on three tabs: **Roof**, **Walls**, and **Base & Floor**. Each tab lists its surfaces on the left. Click one to see its parts on the right.



1. **Tab count.** A red number is how many mismatches that tab has. A green ✓ means none.
2. **Find surface or part…** Type a surface number, a segment, or the number of any part to narrow the list.
3. **Dot.** Red: mismatches. Green: every checked part matches. Grey: nothing on it was checked.
4. **Red count.** How many mismatches this surface has.
5. **Selected surface** (blue row). It is highlighted in the model and its parts fill the right side.

The Walls tab on the example unit. “Segments” tells you which section of the unit the surface belongs to.

Right-click a surface



Copy surface number puts the number on the clipboard as text, ready to paste into an email or a search.

Open in Inventor opens that surface's own assembly in a new Inventor tab. If its file can't be found, you get a message saying so.

Tips

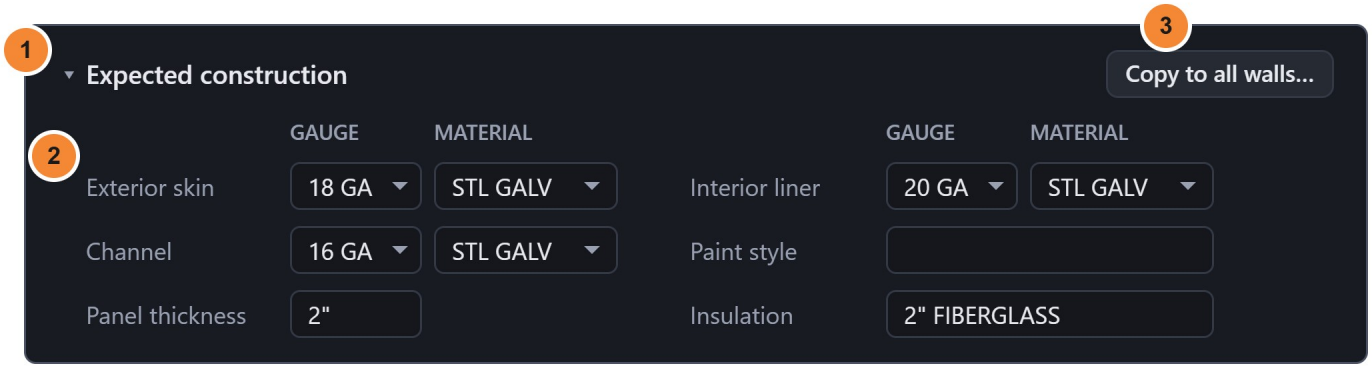
· Clicking a surface brings Inventor back to this unit if you had moved to another file, so the highlight lands on the right model.

· Drag the thin line between the list and the right side to make the list wider or narrower.

· Highlighting also works when you have opened one surface's own assembly directly.

Expected construction

This block shows what the selected surface's parts **should** be, taken from the unit's own configuration. Normally you just read it. Change a value only when you know the order calls for something different.



Expected construction for a wall (example values). Thermal break sits on a further row.

1. Click the heading to fold the block away and give the parts list more room.
2. Pick a different gauge or material from a list. Parts are re-checked at once.
3. **Copy to all walls…** lists which other walls would change and how (for example `Exterior gauge 16 → 18`), then asks. Walls that already match are left alone.

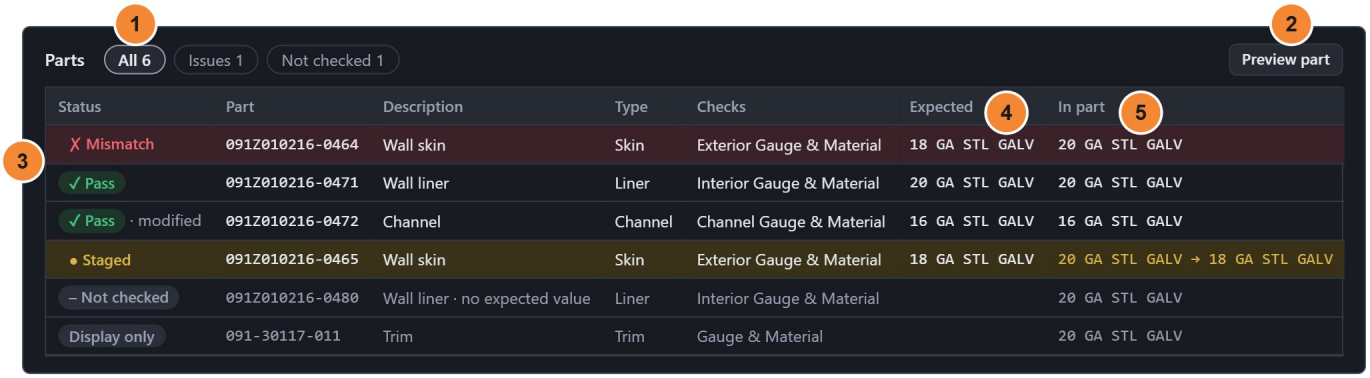
Field	In plain words
Exterior skin	The outside sheet of the panel.
Interior liner	The inside sheet of the panel.
Channel	The formed steel pieces that frame the panel.
Trim (roof only)	Roof caps and roof corner caps. Other trim pieces show as **Display only** and are not checked.
Base & Floor tab	Base height, base structure, formed channel, floor, sub-floor, perimeter angle, floor thermal break. Under **Unit (applies to all bases)**: **Upturned lip** and **Curb rest**.

BE CAREFUL

A changed value shows ● **Unsaved expectation changes** in the footer until you click **Save expectations**. Changing an expectation never changes a part; for that, see chapter 6.

The parts list

Each row is one part on the selected surface. Compare **Expected** (what the order wants) with **In part** (what the model has). The **Status** chip on the left gives the verdict.



Example rows showing every status. In a real list, mismatches sit on a red background and staged changes on a yellow one.

1. **All / Issues / Not checked** filter the rows. Each shows its count.
2. **Preview part** opens a wireframe of the selected part (page 19).
3. **Status**. Hover over it to see why.
4. **Expected**: what the order wants for this check.
5. **In part**: what the part file says now.

Status	It means	What to do
✓ Pass	The part matches.	Nothing.
✗ Mismatch	The part differs from the order.	Decide which is wrong: the part or the expectation (page 12).
• Staged	You queued a change that isn't written yet.	Review and write it, or discard it (chapter 6).
– Not checked	A rule covers it, but it couldn't be checked. The reason follows the description.	Read the reason: no expected value, wrong surface type, or no surface row.
Display only	Listed for information; not checked.	Nothing.

· **modified** after a status means the part was changed after the unit was generated (an SQ part, or edited here). **Right-click** a row for **Copy part number**, **Preview part** and **Open in Inventor**. Parts that no rule covers are not listed at all.

CHAPTER 5

Check a unit, start to finish

This is the everyday routine. It takes a few minutes on a clean unit.

1. Open the unit and click **QA Tools › Verify & Edit Specs** (page 6).
2. Read the result in the header. ✓ All 214 checked parts match means you are done.
3. Otherwise press **F8**. The verifier opens the first surface with a problem, selects its first mismatched part and shows it in the model.
4. Compare **Expected** with **In part** and decide which one is wrong:

   Is the expected value right for this part? Check the order or ask engineering.

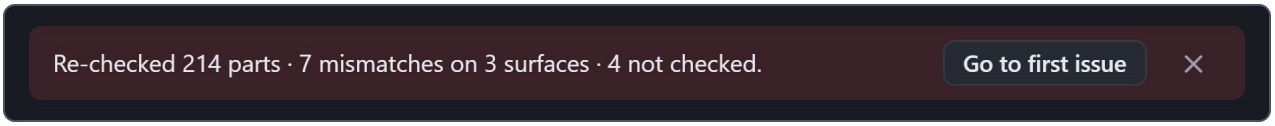
   Yes ↓

   The part is wrong. Fix it with **Edit parts** (chapter 6), or in the model the way you normally would.

   No ↓

   The expectation is wrong. Change it in **Expected construction** (page 10), then **Save expectations**.

5. Press **F8** again for the next problem. Repeat until the result turns green.
6. If you changed anything outside the verifier (in Inventor or YCShell), click **Re-check unit** in the footer. It reads the unit again and shows a summary:

7. If you changed any expectations, click **Save expectations** before closing.

GOOD TO KNOW

The result updates by itself whenever you change something in the verifier, so you don't need **Re-check unit** for that. To share a part number with someone, right-click its row › **Copy part number**, or select the row and press **Ctrl+C**.

CHAPTER 6

Fix parts with Edit parts

When a part has the wrong gauge or material, the verifier can change the part file for you. It works in two stages, so nothing changes by accident: first you **stage** changes (queue them up), then you **review and write** them.

BEFORE YOU START

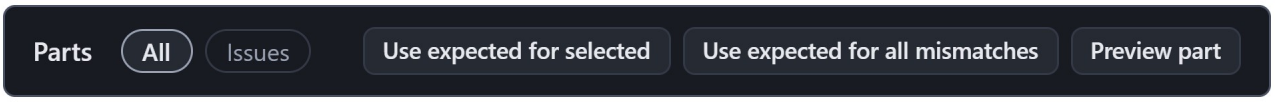
Check the part files out of Vault. Parts that are checked in can't be saved, and their changes will fail.

1 Turn on **Edit parts** in the header. The switch turns yellow and a yellow tray appears in the footer.

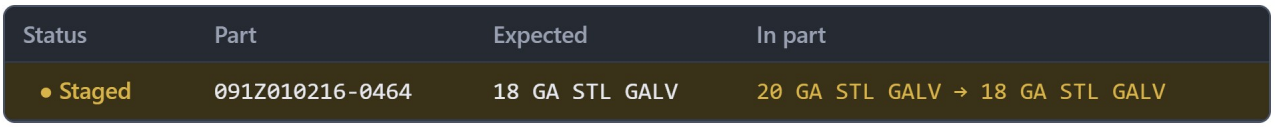


2 Stage the fix in one of three ways:

- Select a row and click **Use expected for selected**.
- Click **Use expected for all mismatches** to fix every mismatch on this surface at once.
- Or double-click the **In part** cell and type the value yourself.



3 The row turns yellow and shows the old and new value. Nothing has been written yet.



4 Check the tray in the footer. It counts what is waiting. **Discard** drops every staged change; **Review & write to parts…** moves on to page 14.

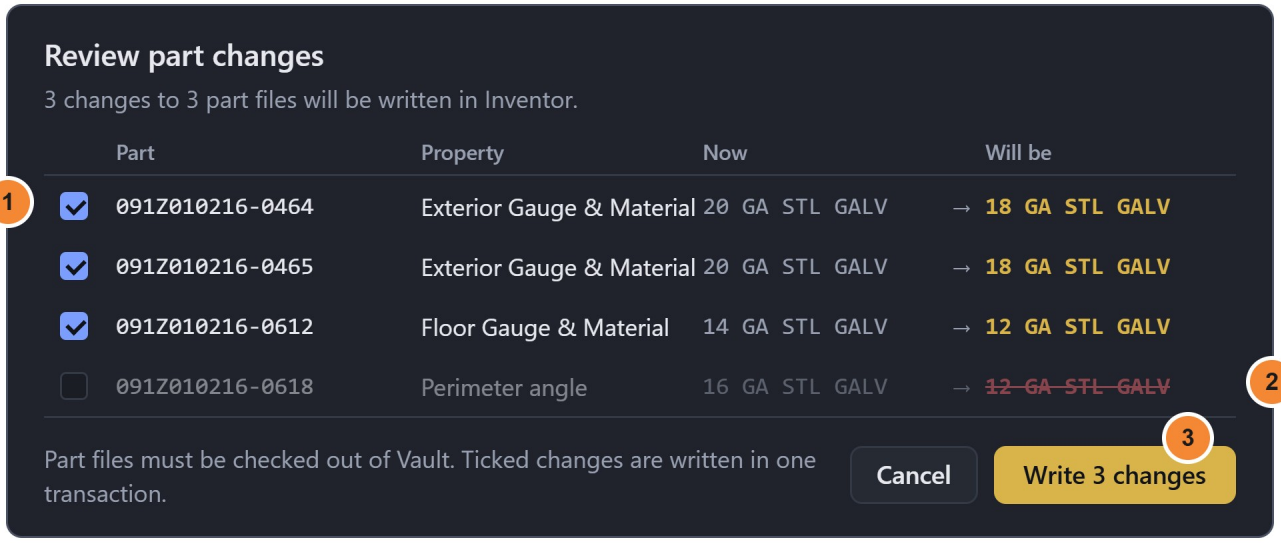


GOOD TO KNOW

Only rows whose check can be written back can be edited: thickness, gauge and material, and base or formed-channel material. Other rows stay read-only. The tray stays visible while changes are staged, even if you turn **Edit parts** off. Closing the verifier never writes staged changes.

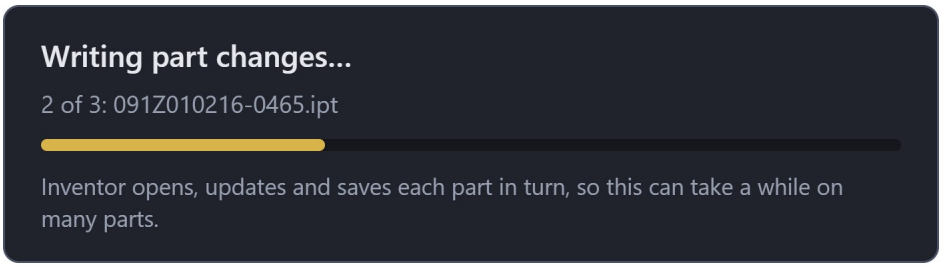
Review and write the changes

5 Click **Review & write to parts…**. A review covers the window and lists every staged change.



1. Untick a change to leave it staged for later.
2. Red and struck through: it can't be written to a part, so it is skipped.
3. **Write 3 changes** writes the ticked ones. **Cancel** writes nothing.

6 Wait while each part is opened, updated and saved. It can take a while on many parts. When it finishes, the verifier re-checks the unit by itself.



What happens to each part

· The new gauge and material are saved in the part file, where MOM and YCShell read them.

· It is marked as a hand-edited (SQ) part, so the next upload carries it to MOM.

· The part turns **yellow** in the model, so anyone can see it was changed by hand.

· Cuts and holes that went through the old sheet are adjusted to go through the new one.

NEXT STEP

The parts are changed in your files only. To send them to MOM, run **Upload changed** (page 16). If you moved or added channels, run **Regenerate changed** first (page 15).

CHAPTER 7

Regenerate and upload changed surfaces

Instead of running YCShell surface by surface, two footer buttons find every surface that needs it and run YCShell for you. YCShell still does the work, exactly as usual.

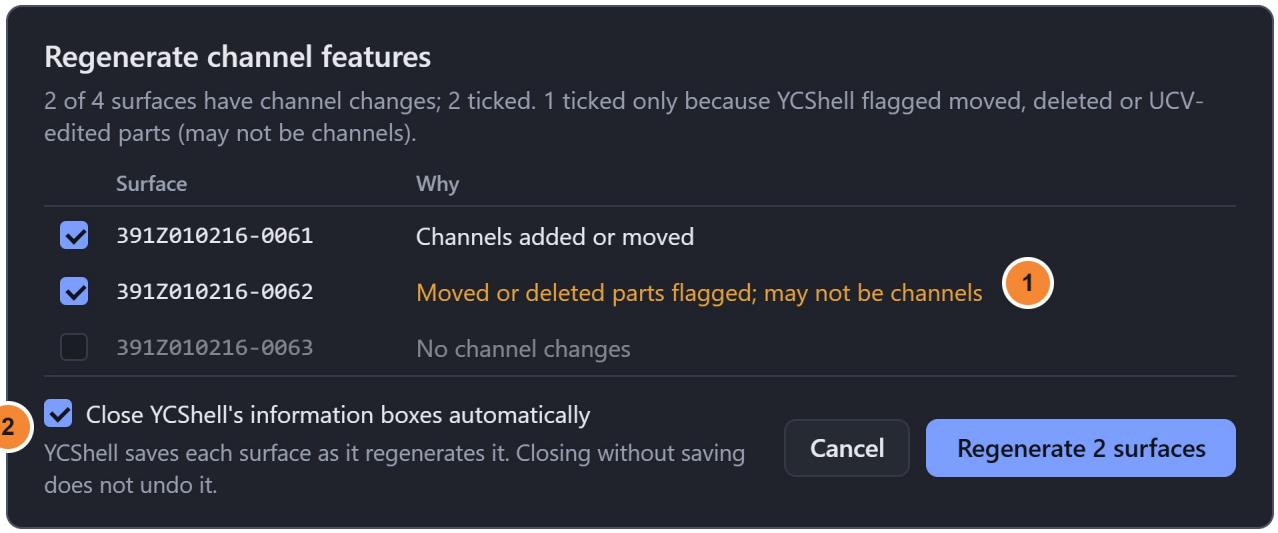
YOU NEED

The JCI MOM (YCShell) add-in loaded and you logged in · the unit open · no staged part edits (write or discard them first).

Regenerate changed

Use it when channels were added (including patterned), reshaped or moved. A skin, liner or gauge change alone does not need it. Bases are never offered; YCShell only regenerates walls and roofs.

1. Click **Regenerate changed (N)…**, where N is how many surfaces need it. A review lists every wall and roof with the reason; those that need it are ticked.



1 Orange: ticked but uncertain; it is safer to regenerate one surface too many. **2** Leave this on; Yes/No questions from YCShell always wait for you. Reason wording is an example.

2. Click **Regenerate N surfaces**. Each surface runs in turn with YCShell's own progress window, about 5 seconds to 2 minutes each. A small window at the top right has **Stop after this surface**. **YCShell saves each surface as it goes**; closing without saving does not undo it.

3. The unit is re-checked and each surface is listed as **Done**, **Done with your answers**, **Not regenerated by YCShell**, **Skipped** (suppressed parts: unsuppress and run again), **Failed** or **Not run**.

Upload changed

Use it when anything on a surface changed since its last upload: a changed or new part (including a gauge fix made in the verifier), an added part such as a patterned channel, or moved or deleted parts. If channels changed too, run **Regenerate changed** first.

1 Click **Upload changed (N)…**. A review lists every surface with its changes.



1 Orange: its channels still need Regenerate. Run that first, or upload anyway. Reason wording is an example.

2 Click **Upload N surfaces**. The verifier first runs YCShell's own checks on each surface (suppressed parts, structural parts with holes). If any fail, nothing is uploaded and you get the full list to fix.

3 **One** YCShell **Upload Changes To MOM** window opens holding every ticked surface. Check it and press **Upload** there, exactly as for one surface. Close it when it says the upload is complete. A **Part #** of **N/A** is normal for a surface MOM renamed. Allow about a minute per surface plus a few minutes overall.

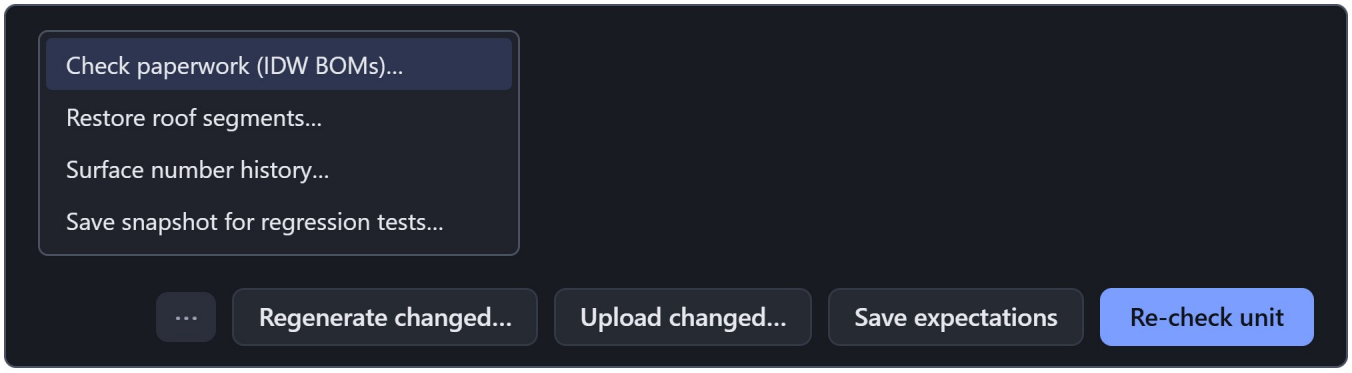
4 The unit is re-checked and each surface gets a result:

Result	What to do
Uploaded	Nothing.
Renamed by MOM and relinked in the unit	Nothing. The unit already points to the new number, for example 0021 → 0048.
Renamed by MOM but not found in the unit	Relink that surface in the unit by hand, then save.
Still changed	It was not uploaded or the upload failed. Read YCShell's upload log in `%LOCALAPPDATA%\Autodesk\ISG\MOM_YCShell\Logs`

CHAPTER 8

More tools: paperwork, history, preview

The ⋯ button at the left of the footer buttons opens four extra tools. **Restore roof segments** has its own page (18); the other three don't change your model.



Check paperwork (IDW BOMs)

Reads each surface's paperwork drawing (its `.idw`) and lists bill-of-materials rows whose printed gauge or material differs from the part as modelled. It only reports; nothing is changed. The report is also copied to the clipboard.

```
391Z010216-0061: 2 of 4 rows differ (from 391Z010216-0061.idw)
   Liners:2 M 091Z010216-0464: IDW 22 GA STL GALV, part 20 GA STL GALV
```

How to read a line: the drawing (IDW) prints 22 gauge for this liner, but the part in the model is 20 gauge.

· The drawing used is the one next to the surface, or, if there is none, the one inside the surface's `.ZIP` from its last upload. Either way it is the paperwork MOM received.

· YCShell prints material from the **surface configuration**, not the part. If you changed a part away from it, the paper keeps the old value: fix the surface configuration in YCShell, then upload again.

· Channel and insulation rows print no material and are skipped. A row whose part isn't in the unit is listed as **part not found**.

Surface number history

MOM sometimes gives a surface a new number when it is uploaded. The verifier follows it, moving its expectations along. This tool lists every number each surface has had, with dates, for example `0021 → 0048 → 0049 → 0122`, and copies the list to the clipboard.

Save snapshot for regression tests

For the people who maintain the verifier. It saves what the verifier read from a unit known to be correct, so later versions can be tested against it. Everyday users can ignore it.

Restore roof segments

Use this after you raise a sloped roof's **YLength** so its slope lands on the next roof. MOM then rebuilds that roof without its **segment**, so YCShell no longer knows what it is made of: skins, liners and trim come out as `20 GA STL GALV`, the foam is empty or printed as Foamboard (**FB**) instead of injected foam (**IF**), and the paperwork BOM prints **Error getting segment and material.**

Before you start: open the whole unit, not just the roof (the verifier copies the roof's segment from the other surfaces). Load the JCI MOM (YCShell) add-in, log in, and load the roof in YCShell.

1. Click ··· › **Restore roof segments…** in the footer.
2. Read the review. Each roof that needs fixing is listed with what is added (**+**), removed (**−**) and changed (**~**). Click **Yes** to restore them.

```
391Z010216-0070  (roof top 133.79 is above its segments' top 133 (YLength raised))
   + VESTIBULE: skin 16 GA STL GALV, liner 18 GA ALM SHT, insulation Foam
   ~ nominal thickness 2.79" → 2" (the segment's panel thickness), so YCShell finds its 2" foam
spec instead of making the foam Foamboard
```

An example review. The roof's shape is not changed; only its segment and the thickness YCShell uses for foam.

3. Read the result. Each roof gets one line, for example `391Z010216-0070: restored VESTIBULE, nominal thickness 2.79" → 2"; insulation: 7 Foam`. **Foam** is right. **FoamBoard** means YCShell found no foam for the roof: tell the maintainer.
4. Close the verifier and open it again on the unit. The roof's expected skin and liner now come from the order; fix any parts it flags (chapter 6). Then in YCShell, **Generate Paperwork** and check the insulation page shows **IF**, and **Upload changed** (page 16).

BE CAREFUL

This changes the roof's configuration and saves the roof, nothing else. There is no undo button; the old configuration is kept in the `.ucv_config_backups` folder next to the roof. If the result starts with **RESTORE FAILED**, read that roof's line. **Failed part way** means the roof may be changed but not saved: don't save, close the unit without saving, reopen it, and send the log (page 20).

NOT PUT BACK

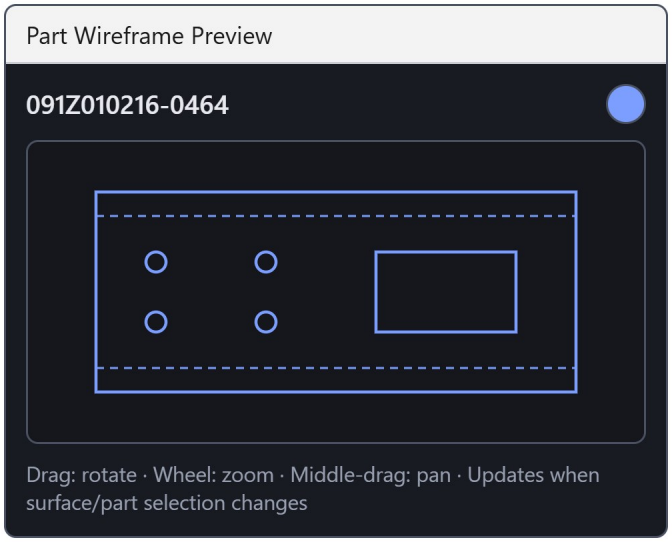
Openings on the roof: the review counts them, and you add them by hand. Bulkhead records. A roof whose parts didn't change is skipped by **Upload**.

Preview, inspect, save

Preview part

Select a part, then click **Preview part** (or right-click › **Preview part**). A wireframe opens beside the verifier showing the flat sheet with its cuts and holes. It follows your selection as you click other parts.

Drag to rotate, **scroll** to zoom, **middle-drag** to pan. The dot at the top right sets the line colour.



Thickness Inspector

Click **QA Tools › Thickness Inspector** and rest the mouse on any part to see its gauge, material and recorded thickness. Click again or press **Esc** to stop.

· **Measured 0.0598"**: the model's real thickness differs from the recorded one.

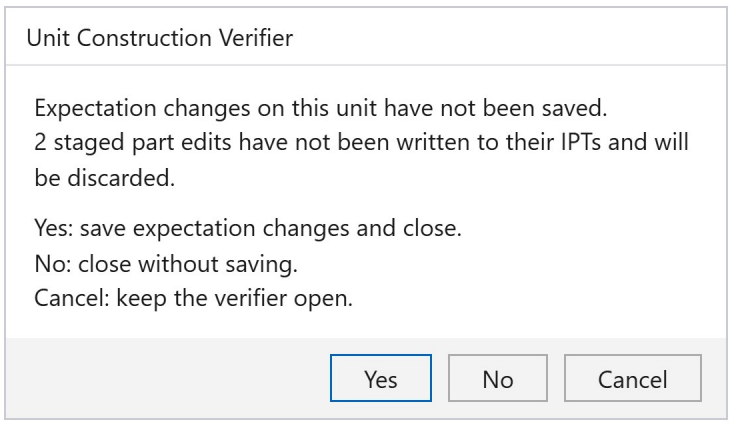
· **No gauge match**: the gauge isn't in the materials list.

· **No thickness recorded**: nothing in the part says how thick it is.



Saving and closing

Save expectations keeps your expectation changes for this unit in a small file next to it. If you close the verifier with something unsaved, it asks first:



BE CAREFUL

Staged part edits are never written on close; write them first if you want them. Closing **Inventor** while the verifier is docked does **not** ask, so click **Save expectations** first.

CHAPTER 9

When something goes wrong

What you see	Try this
Installer shows **[WARNING] Autodesk Inventor is currently running.**	Save your work, close Inventor completely, then press any key in the black window to carry on.
Installer shows **[ERROR] Installation failed!**	Close Inventor and run it again from the extracted folder. If it still fails, contact the verifier's maintainer.
No **QA Tools** tab, or no **Verify & Edit Specs** button	Open an assembly (the button only shows there). Turn the add-in on in **Tools › Add-Ins** (page 5, step 6). If it isn't listed, install again (page 4).
Result says **Not scanned**	Open the unit's main assembly, not a single surface, and open the verifier again.
Nothing is highlighted in the model	Set **Highlight** to **Box** or **X-Ray**, then click the surface again.
Writing a part fails	Check the part out of Vault. Only sheet metal parts can be written.
A change is red and struck through in the review	That value can't be written to a part. Change it in Inventor instead.
Regenerate or Upload says YCShell can't be reached	Load the JCI MOM add-in and log in to the MOM tools, then try again.
YCShell version not supported	YCShell was updated. These buttons stay off until the verifier is updated; tell its maintainer.
A surface you changed is not offered	Save the surface, click **Re-check unit** and try again. If it is still missing, run YCShell on it by hand and tell the maintainer.
Perimeter angles mismatch on a base whose lip runs on only some sides	Expected. The standard angles on the sides without a lip show as mismatches by design.
Inventor pastes an old part or asks about missing references	Click **Cancel** in Inventor. See “Keys and the clipboard” on page 21.

ASKING FOR HELP

Send the unit name, the version (1.0.40, or the one on your zip), what you clicked, a screenshot, and the log file `%TEMP%\UnitConstructionVerifier\ucv_debug.txt` . Paste that path into File Explorer's address bar to find it.

CHAPTER 10

Words to know and quick reference

Unit · the whole air handler, one Inventor assembly (.iam).

Surface · a roof panel, wall, base or floor; its own sub-assembly.

Segment · the section of the unit a surface belongs to.

Part · one sheet metal piece (.ipt): skin, liner, channel, trim.

Expected · what the order says a part should be.

In part · what the part in the model actually is.

Mismatch · expected and in part differ.

Re-check · read the unit from Inventor again and redo every check.

Expectations · the expected values for a unit, saved next to it.

Gauge (GA) · sheet thickness number. A smaller number is a thicker sheet.

STL GALV · galvanised steel, as written on parts and drawings.

Staged · a part change queued in the verifier, not yet written.

Write · save staged changes into the part files.

SQ part · a part changed by hand after the unit was generated.

YCShell · the JCI MOM add-in that regenerates channels and uploads surfaces.

MOM · the system surfaces are uploaded to; it produces the paperwork.

IDW · an Inventor drawing; here, a surface's paperwork with its parts list.

Vault · where part files are checked out before editing and back in after.

Keys

Press	To
F8	Go to the next surface with a mismatch and select its first bad part.
Ctrl+C	Copy the selected part or surface number as text.
Double-click an In part cell	Type a new value (only with **Edit parts** on).
Ctrl+V	Paste into a text box only, so an old part can't land in the model. Back in Inventor, it pastes whatever Inventor last copied.
Esc	Stop the Thickness Inspector.

While the verifier is active, keys go to it and Inventor's own shortcuts don't fire; **Alt** shortcuts still reach Inventor.

Files kept next to a unit

Saved expectations, `.ucv_channel_fingerprints.json` and `.ucv_surface_history.json`; next to a restored roof, the `.ucv_config_backups` folder (page 18). Leave them with the unit; deleting them loses saved expectations and surface history.